8.2 Test Planning and Estimation

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8.2 Test planning and estimation

- Test Planning
- Test Planning Activities
- Exit Criteria
- Test Estimation
- Test Approaches

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Test Planning

- Planning may be documented in a master test plan and in separate test plans for test levels such as system testing and acceptance testing.
- Test planning is a continuous activity and is performed in all life cycle processes and activities.



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Purpose and Content of Test Plan

- A test plan outlines test activities for development and maintenance projects.
- As the project and test planning progress, more information becomes available and more detail can be included in the test plan.



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Purpose and Content of Test Plan (cont.)

- Test planning is a continuous activity and is performed throughout the product's lifecycle
- Feedback from test activities should be used to recognize changing risks so that planning can be adjusted
- Planning may be documented in a master test plan and in separate test plans for test levels or for separate test types.

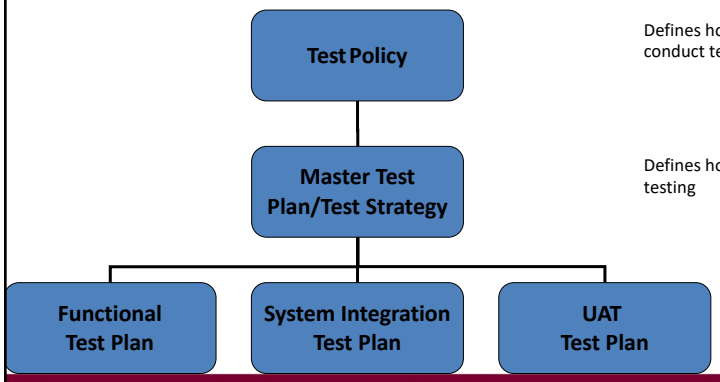


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Test Planning

- All projects require a set of plans and strategies which define how the testing will be conducted.
- There are number of levels at which these are defined:



Defines how the **organisation** will conduct testing

Defines how the **project** will conduct testing

Defines how each **level** of testing will be conducted

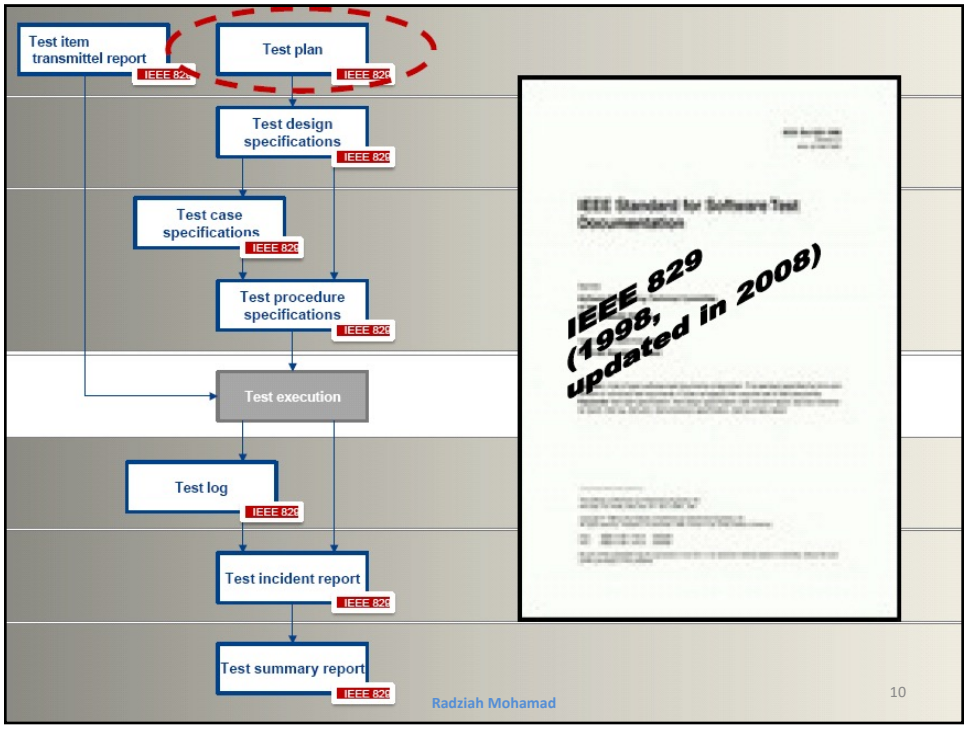
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Master Test Plan - Structure

a) Test plan identifier;
b) Introduction;
c) Test items;
d) Features to be tested;
e) Features not to be tested;
f) Approach;
g) Item pass/fail criteria;
h) Suspension criteria and resumption requirements;
i) Test deliverables;
j) Testing tasks;
k) Test environment needs;
l) Responsibilities;
m) Staffing and training needs;
n) Schedule;
o) Risks and contingencies;
p) Approval procedures.

IEEE Std 829-1998
IEEE Standard for Software Test Documentation "Test Plan"

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Test Planning Activities

- Determine scope, risks, and objectives of testing
- Define the approach of testing (definition of the test levels/entry and exit criteria)
- Integrate & coordinate test activities into the software life cycle activities
- Decide what to test, and how
- Select metrics
- Define the amount of documentation required
- Assign resources



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Entry Criteria

- Entry criteria (definition of ready) define the preconditions for undertaking a given test activity.
- Typical entry criteria include:
  - Availability of testable requirements, user stories, and/or models
  - Availability of test items that have met the exit criteria for any prior test levels
  - Availability of test environment
  - Availability of necessary test tools
  - Availability of test data and other necessary resources



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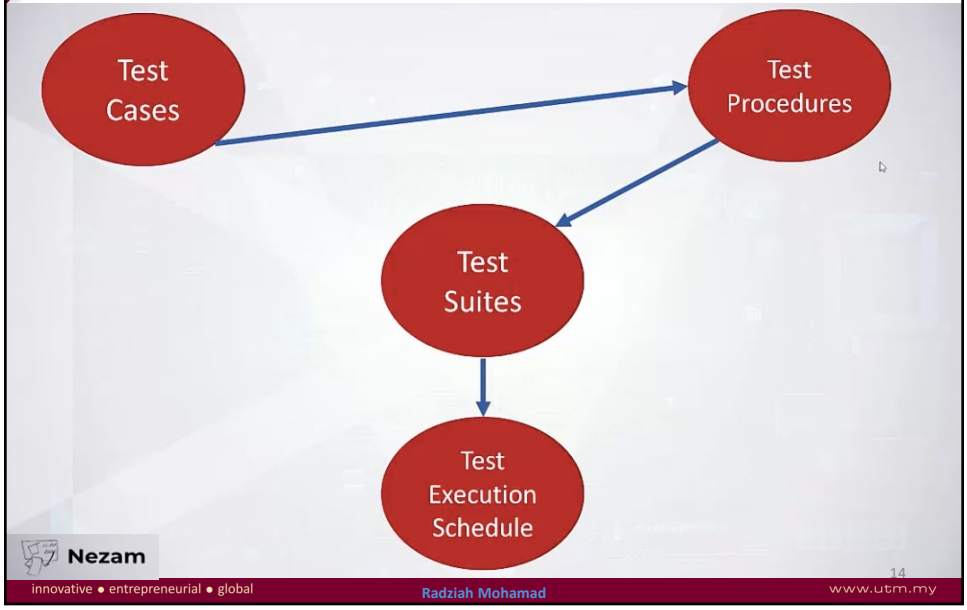
Exit Criteria

- Exit criteria (definition of done) define what conditions must be achieved in order to declare a test level or a set of tests completed.
- Typical exit criteria include:
  - Planned tests have been executed
  - A defined level of coverage has been achieved
  - The number of unresolved defects is within an agreed limit
  - The number of estimated remaining defects is sufficiently low
  - The evaluated levels of quality characteristics are sufficient



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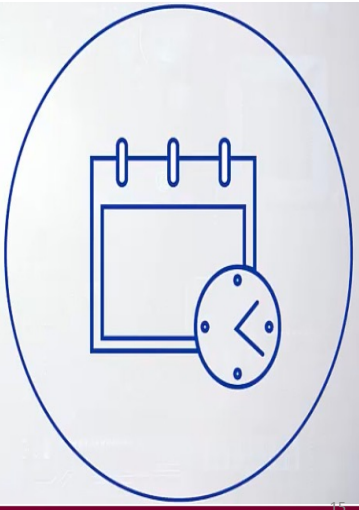


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Test Execution Schedule

- Ideally, test cases would be ordered to run based on their priority levels
- If a test case with a higher priority is dependent on a test case with a lower priority, the lower priority test case must be executed first

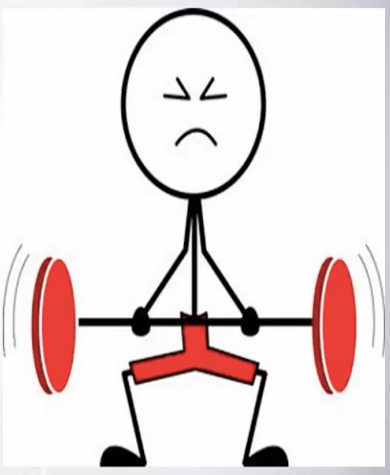


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Factors Influencing Test Efforts

1. Product Characteristics
   - Risks
   - Quality of the test basis
   - Size-Complexity-Requirements
   - Documentation Required



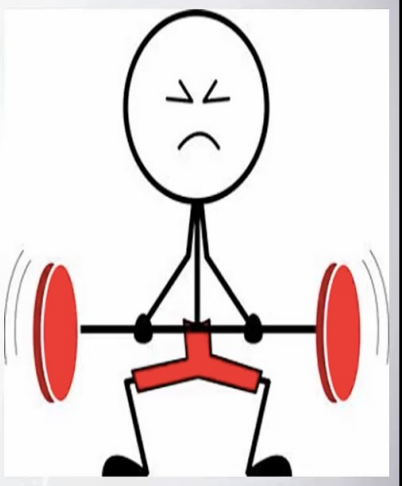
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Factors Influencing Test Efforts

2. Development Process Characteristics

- The stability and maturity of the organization
- The development model in use
- The test approach
- The tools used
- The test process
- Time pressure



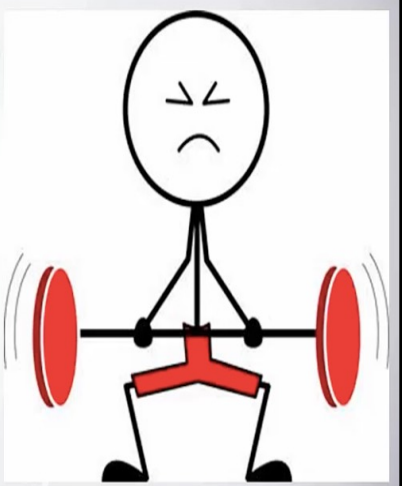
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Factors Influencing Test Efforts

3. People Characteristics

- The skills and experience of the people involved, especially with similar projects and products
- Team cohesion and leadership



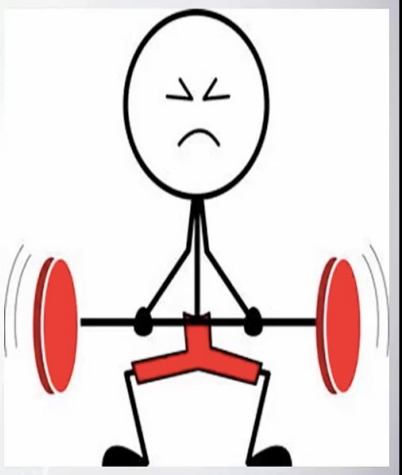
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Factors Influencing Test Efforts

4. **Test Results**

- The number and severity of defects found
- The amount of rework required



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Test Estimation Techniques

1. **Metrics-based**

- estimating the test effort based on metrics of former similar projects, or based on typical values

2. **Expert-based**

- estimating the test effort based on the experience of the owners of the testing tasks or by experts



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Metrics-Based Example

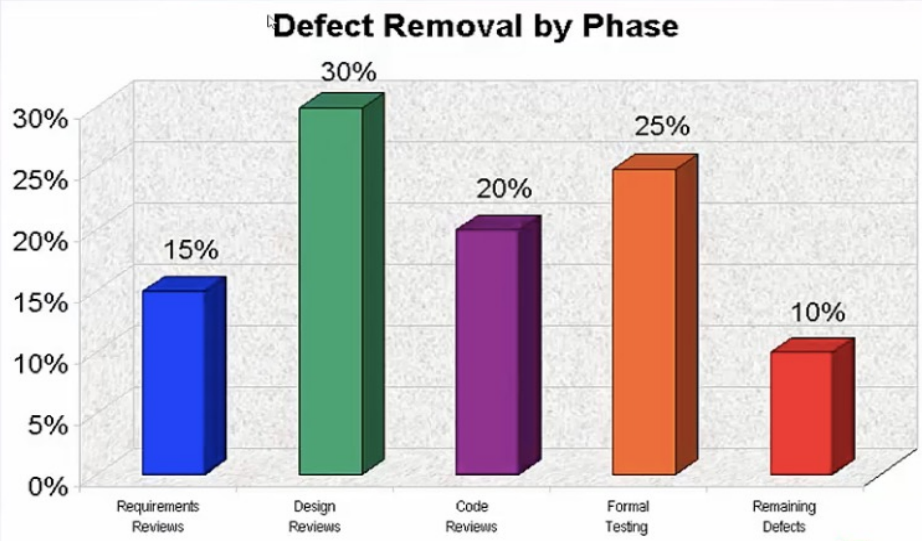


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Agile/Metrics-based

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Metrics-Based Example



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Expert-Based Example



Agile/Expert-based

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Expert-Based Examples

Estimator: _____ Date: _____ Page: _____

Activity: Develop requirements for Project X Estimation Units: labor hours

Category: ___ Project Tasks ___ Quality and Process Tasks ___ Waiting Time

Task	Initial Estimate	Change #1	Change #2	Change #3	Change #4	Final
Identify user representatives	3	-1				
Hold requirements workshops	16	+8				
Draft requirements specification	16	+3				
Review requirements specification	8	+4				
Develop prototype	24	-8				
Correct errors following reviews		+5				
Net Change		+10				
Total	67	78				

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Figure 2: Sample Wideband Delphi estimation form.

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